MATHEMATICS
Modern Algebra I
Solutions to excercises from chapter 3

1) Let $f : U \oplus W \to U + W$ be the homomorphism such that $f(u, v) = u + v$. Clearly f is surjective and $ker(f) = \{(v, -v)\ s.t.\ v \in U \cap W\}$. By Theorem III.5.3

$$dimU + dimW = dim(U \oplus W) =$$
$$dimker(f) + dimIm(f) = dim(U \cap W) + dim(U + W).$$

3) Let $dim_k R = n$. Pick $x \in R$ then $1, x, x^2, ..., x^n$ are linearly dependent over k and so there is a linear combination $c_0 + c_1 x + ... + c_n x^n = 0$. Not all c_i are zero, so fix r the smallest number such that $c_r \neq 0$. Then $c_r x^r + ... + c_n x^n = 0$, and so $x^r(c_r + c_{r+1}x + ... + c_n x^{n-r}) = 0$. If $x \neq 0$, since R is entire, one has $c_r + c_{r+1}x + ... + c_n x^{n-r} = 0$ and so $1 = x(-c_{r+1} - c_{r+2}x - ... - c_n x^{n-r-1})c_r^{-1}$, i.e. x is invertible for any $x \in R - 0$.

4 b) Let $\Psi : E \to \oplus E_i$ with $\Psi(x) = (\psi_1(x), ..., \psi_m(x))$ and $\Phi : \oplus E_i \to E$ with $\Phi(x_1, ..., x_m) = \phi_1(x_1) + ... + \phi_m(x_m)$. One easily checks that these are homomorphisms of modules. To see that they are isomorphisms, we show that $\Psi \circ \Phi = id$ and $\Phi \circ \Psi = id$.

$$\Psi \circ \Phi(x_1, ..., x_m) = \Psi(\phi_1(x_1) + ... + \phi_m(x_m)) =$$
$$(\psi_1(\sum \phi_i(x_i)), ..., \psi_m(\sum \phi_i(x_i))) =^{1}$$
$$(\psi_1(\phi_1(x_1)), ..., \psi_m(\phi_m(x_m))) =^{2} (x_1, ..., x_m).$$

$$\Phi \circ \Psi(x) = \Phi(\psi_1(x), ..., \psi_m(x)) = \phi_1(\psi_1(x)) + ... + \phi_m(\psi_m x)) =$$
$$(\sum \phi_i \circ \psi_i)(x) = x.$$

If $E \cong \oplus E_i$, then the above properties are clear.

9) a) Define $S^{-1}A \times S^{-1}M \to S^{-1}M$ by $(a/s) \cdot (m/s') = (a \cdot m/ss')$ where $a \cdot m$ is given by the A-module structure of M. It is an easy check to see that this makes $S^{-1}M$ in to an $S^{-1}A$ module eg.

$$(\frac{a_1}{s_1} + \frac{a_2}{s_2})\frac{m}{s'} = \frac{a_1 s_2 + a_2 s_1}{s_1 s_2}\frac{m}{s'} =$$

[1] as $\psi_i \circ \phi_j = 0$ for $i \neq j$
[2] as $\psi_i \circ \phi_i = id$

$$\frac{(a_1s_2+a_2s_1)m}{s_1s_2s'}=\frac{a_1s_2m+a_2s_1m}{s_1s_2s'}=\frac{a_1s_2m}{s_1s_2s'}+\frac{a_2s_1m}{s_1s_2s'}=$$
$$\frac{a_1m}{s_1s'}+\frac{a_2m}{s_2s'}=(\frac{a_1}{s_1})\frac{m}{s'}+(\frac{a_2}{s_2})\frac{m}{s'}.$$

b) Let $f : M' \to M$ and $g : M \to M''$ and $F : S^{-1}M' \to S^{-1}M$ and $G : S^{-1}M \to S^{-1}M''$ the corresponding maps. Note $F(m'/s) = f(m')/s$ and $G(m/s) = g(m)/s$.

Claim: F is injective. Suppose $F(m'/s) = 0$, then $f(m')/s = 0$ then $tf(m') = 0$ for some $t \in S$. So $f(tm') = 0$ and since f is injective, $tm' = 0$ and hence $m'/s = (tm')/(ts) = 0$.

Claim: G is surjective. Let $m''/s \in S^{-1}M''$, then $m'' = g(m)$ so $m''/s = G(m/s)$.

Claim: $Im(F) \subset Ker(G)$ i.e. $G \circ F = 0$. $G \circ F(m'/s) = (g \circ f)(m')/s = 0/s$.

Claim: $)Ker(G) \subset Im(F)$. $G(m/s) = 0$ implies $g(m)/s = 0$ implies $tg(m) = 0$ for some $t \in S$ and so $g(tm) = 0$ and so $tm = f(m')$ and so $m/s = tm/ts = F(m'/ts)$.

14) a) follows from the snake lemma as

$$Ker(f) \to Ker(g) \to Ker(h)$$

is exact and so $Ker(f) = 0$, $Ker(h) = 0$ implies $Ker(g) = 0$.

b) is analogous to a).

c) If f, h are isomorphisms, then by a), b) so is g.

If $0 \to M' \to M$ is exact and g, h are isomorphisms, then by the snake lemma

$$Ker(h) \to Coker(f) \to Coker(g)$$

is exact. But $0 = Coker(g)$, $0 = Ker(h)$ and so $0 = Coker(f)$ i.e. f is injective. Since $M' \to M$ is injective, one sees that $Ker(f) \to Ker(g) = 0$ is also injective and so $Ker(f) = 0$ and so f is also injective i.e. f is an isomorphism.

The last statement is similar.

17) The maps $f_{n,m} : A_n \to A_m$ for $n \geq m$ are just $x+p^n\mathbb{Z} \to x+p^m\mathbb{Z}$. It is easy to see that these are well defined homomorphism. Clearly $f_{i,i} = id$ and $f_{m,p} \circ f_{n,m} = f_{n,p}$ for $n \geq m \geq p$.

Claim: $\mathbb{Z}_p \to \mathbb{Z}/p^n\mathbb{Z}$ is surjective. Let $x_n = x + p^n\mathbb{Z} \in p^n\mathbb{Z}$, then define $x_i = x + p^i\mathbb{Z} \in A_i$. One sees that $f_{i,j}(x_i) = x_j$ for $i \geq j$ and so $\{x_i\}$ is an element of $\mathbb{Z}_p$ which maps to x_n in $\mathbb{Z}/p^n\mathbb{Z}$.

One has $\{x_i\} \cdot \{y_i\} = \{x_i + y_i\}$ and $\{x_i\} \cdot \{y_j\} = \{x_k y_k\}$.

Claim: There are no zero divisors. Suppose $\{x_i\} \cdot \{y_j\} = 0$, then $x_i y_i = 0$ for all i. If $\{x_i\} \neq 0$, then $x_i \neq 0$ for some $i = i_0$ and hence for all $i \geq i_0$, similarly $y_j \neq 0$ for all $j \geq j_0$. So for $k \geq i_0 + j_0 - 1$, one has $x_k y_k \neq 0$ and so $\{x_i\} \cdot \{y_j\} \neq 0$.[3]

Claim: Every element $\{x_i\} \in \mathbb{Z}_p - p\mathbb{Z}_p$ is invertible (and therefore $p\mathbb{Z}_p$ is the unique maximal ideal).

We have $x_0 \neq 0 \in \mathbb{Z}/p\mathbb{Z}$ and so $x_i \in \mathbb{Z}/p^i\mathbb{Z} - p\mathbb{Z}/p^i\mathbb{Z}$ are all invertible. Define $\{y_i\}$ by $y_i = x_i^{-1}$. Then $f_{i,j}(y_i) = f_{i,j}(x_i^{-1}) = f_{i,j}(x_i)^{-1} = (x_j)^{-1} = y_j$ and so $\{y_j\} \in \mathbb{Z}_p$. It is easy to see that $\{x_i\} \cdot \{y_j\} = 1$.

Similarly one sees that any element $x \in \mathbb{Z}_p$ is of the form $unit \cdot p^t$ and so p is the only prime in $\mathbb{Z}_p$ and $\mathbb{Z}_p$ is factorial.

b) There is an obvious map $R := lim\mathbb{Z}/(a) \to lim\mathbb{Z}/(p) = \mathbb{Z}_p$. Namely, an element $x \in R$ is a "compatible" collection $\{x_a\}$ with $x_a \in \mathbb{Z}/(a)$. Clearly, one can consider only the subset of $a = p^s$, and obtain a "compatible" collection $\{x_{p^s}\}$ with $x_{p^s} \in \mathbb{Z}/(p^s)$. This is an element of $\mathbb{Z}_p$. So we get a homomorphism

$$\Psi : R \to \prod \mathbb{Z}_p.$$

Claim: Ψ is injective. Suppose that $\Psi(\{x_a\}) = 0$, then $x_{p^s} = 0$ for all primes p and all positive integers s. For any $a = p_1^{s_1} ... p_t^{s_t}$ one has that $x_a \in \mathbb{Z}/(a) \cong \mathbb{Z}/(p_1^{s_1}) \times ... \times \mathbb{Z}/(p_t^{s_t})$. But the image of x_a in $\mathbb{Z}/(p_j^{s_j})$ is just $x_{p_j} = 0$ and hence $x_a = 0$ and so $\{x_a\} = 0$ as required.

Claim: Ψ is surjective. Let $\{x_{p,*}\} \in \prod \mathbb{Z}_p$ so that $x_{p,*} \in \mathbb{Z}_p$. Write $x_{p,*} = \{x_{p,n}\}$ with $x_{p,n} \in \mathbb{Z}/(p^n)$. For any positive integer a write $a = p_1^{s_1} ... p_t^{s_t}$ and define x_a by using the isomorphism of the preceding claim. One checks that the sequence $\{x_a\}$ defines an element of $lim\mathbb{Z}/(a)$ such that $\psi(\{x_a\}) = \{x_{p,*}\}$.

23) We must show that

$$limHom(N, M_i) = Hom(N, limM_i).$$

Let $f_{i,j} : M_i \to M_j$ for $i \geq j$ be the homomorphisms defining the projective system M_i. An element $\psi \in limHom(N, M_i)$ is a collection of homomorphisms $\{\psi_i : N \to M_i\}$ such that $f_{i,j} \circ \psi_i = \psi_j$ for $i \geq j$.

We define a homomorphism $f : limHom(N, M_i) \to Hom(N, limM_i)$ by $f(\psi) = \phi$ where $\phi(n) = f(\psi)(n) := \{\psi_i(n)\}$. We must check that $\{\psi_i(n)\} \in limM_i$ i.e. that $f_{i,j}(\psi_i(n)) = \psi_j(n)$. This is true as $f_{i,j} \circ \psi_i = \psi_j$.

We define a homomorphism $g : Hom(N, limM_i) \to limHom(N, M_i)$ as follows: for an element $\phi \in Hom(N, limM_i)$, let $g(\phi) = \{\psi_i\}$ where $\phi(n) = \{m_i\} =: \{\psi_i(n)\}$ (i.e. $\psi_i(n) := m_i$). We must check that $f_{i,j} \circ \psi_i = \psi_j$, but $f_{i,j} \circ \psi_i(n) = f_{i,j}(m_i) = m_j = \psi_j(n)$.

[3] p^{i_0} does not divide x_i for any $i \geq i_0$ and p^{j_0} does not divide y_j for any $j \geq j_0$. Therefore, $p^{i_0+j_0-1}$ does not divide $x_k y_k$ for any $k \geq i_0 + j_0 - 1$

It is easy to see that f, g are inverses.

27) $R = \oplus R_i = \oplus A_i/A_{i-1}$. For $x \in R$, write $x = (x_0, x_1 + A_0, x_2 + A_1, x_3 + A_2, ...)$ where all but finitely many $x_i + A_{i-1}$ are nonzero. Define $x \cdot y$ as

$$(x_0, x_1 + A_0, x_2 + A_1, x_3 + A_2, ...) \cdot (y_0, y_1 + A_0, y_2 + A_1, y_3 + A_2, ...) =$$
$$(x_0y_0, x_0y_1 + x_1y_0 + A_1, x_0y_2 + x_1y_1 + x_2y_0 + A_1, ...)$$

i.e. the i-th term is given by $\sum x_{i-k}y_k + A_{i-1}$. We must check that this is well defined. Suppose that $(x_0, x_1 + A_0, x_2 + A_1, x_3 + A_2, ...) = x'_0, x'_1 + A_0, x'_2 + A_1, x_3 + A_2, ...)$ i.e. $x'_i - x_i = a_{i-1} \in A_{i-1}$, then we must check that $\sum x_{i-k}y_k + A_{i-1} = \sum x'_{i-k}y_k + A_{i-1}$. The point is that $x_{i-k}y_k - x'_{i-k}y_k + A_{i-1} = (x_{i-k} - x'_{i-k})y_k + A_{i-1} = a_{i-k}y_k + A_{i-1} = 0 + A_{i-1}$.

We must now check that $R_iR_j \subset R_{i+j}$. So we compute

$$(0, ..., 0+A_{i-2}, x_i+A_{i-1}, 0+A_i, ...)\cdot(0, ..., 0+A_{j-2}, y_j+A_{j-1}, 0+A_j, ...)$$
$$= (0, 0 + A_0, ..., 0 + A_{i+j-2}, x_iy_j + A_{i+j-1}, 0 + A_{i+j}, ...) \in R_{i+j}.$$